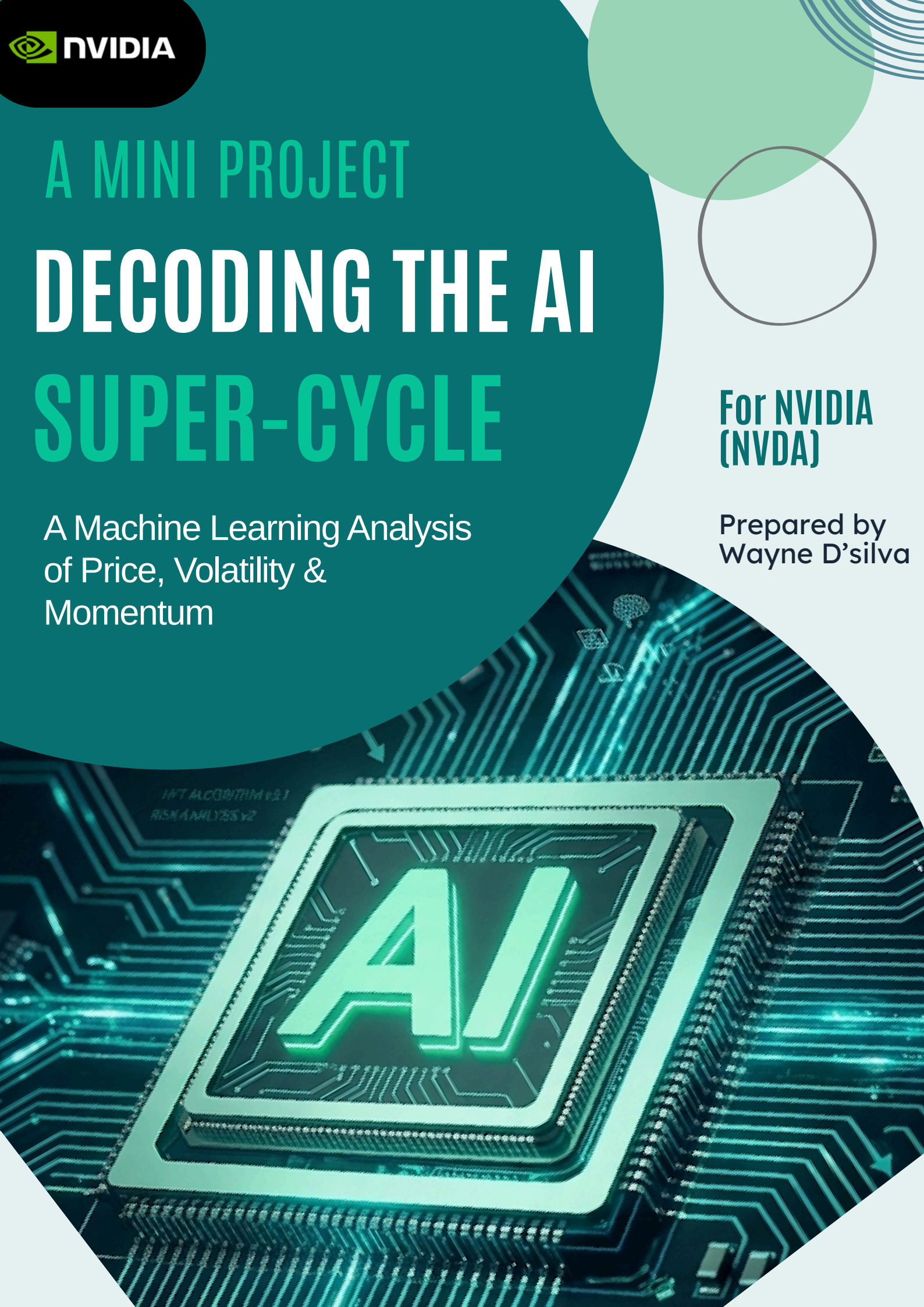




A MINI PROJECT

DECODING THE AI SUPER-CYCLE

A Machine Learning Analysis
of Price, Volatility &
Momentum

For NVIDIA
(NVDA)

Prepared by
Wayne D'silva

MARKET REGIME DETECTION

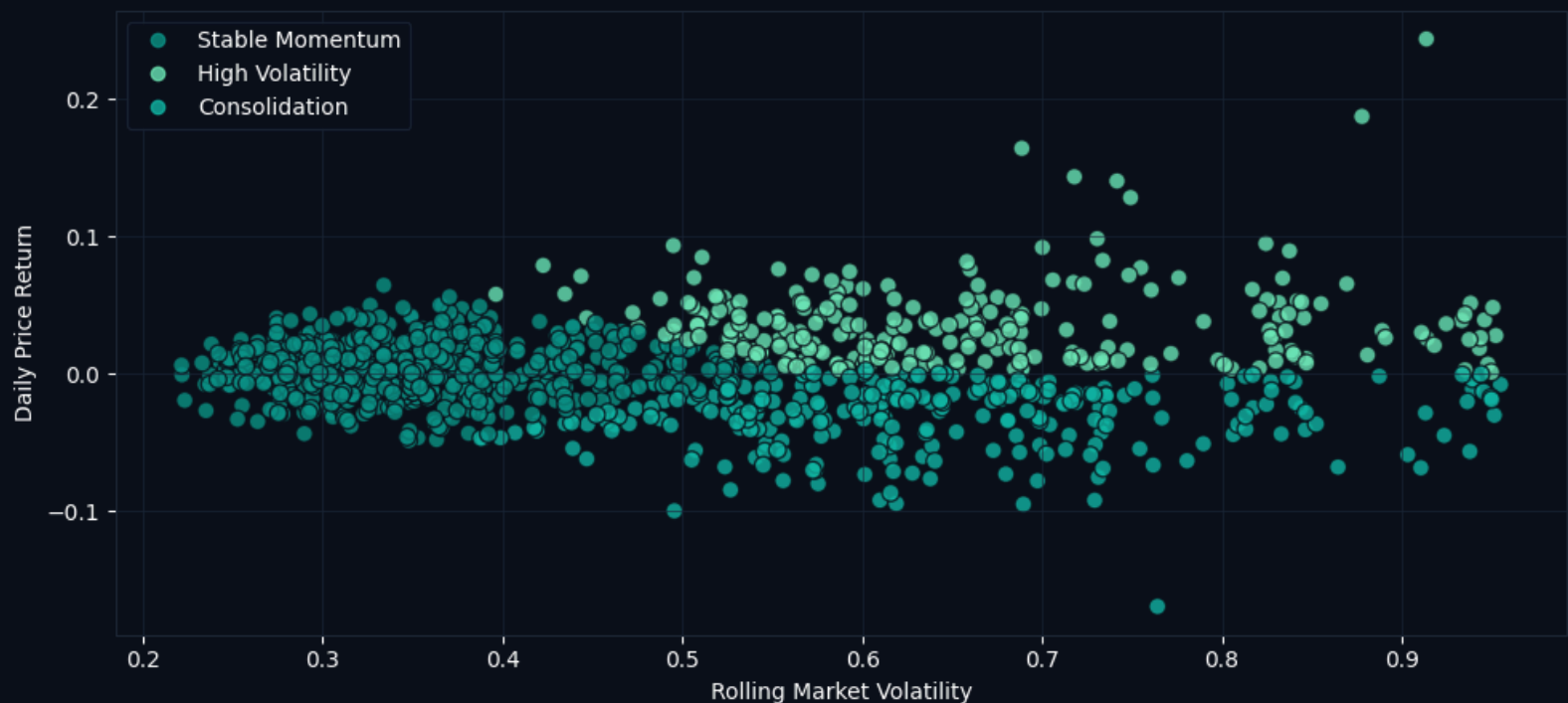


1,000+ TRADING DAYS ANALYZED

NVDA: Historical Price Action



K-Means: Market Behavior Regimes



K-Means clustering segmented NVDA into:

- Stable Momentum
- High Volatility
- Consolidation

Regression: Next-Day Price Prediction



PRICE FORECASTING

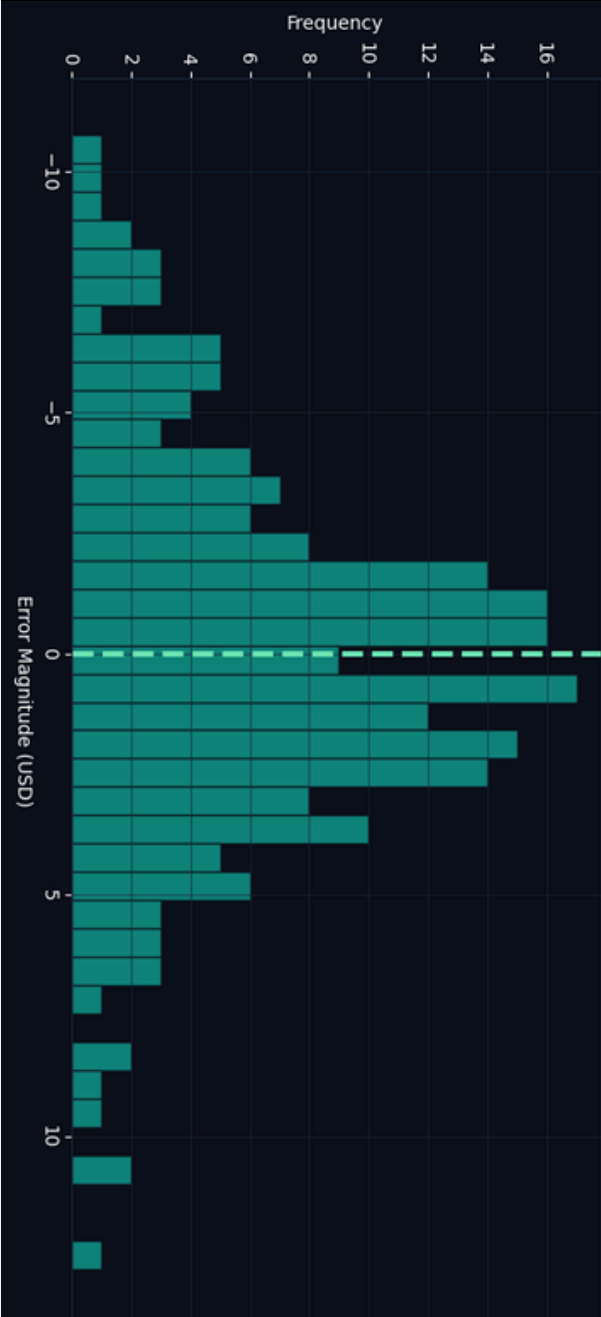
86.6%

REGRESSION
ACCURACY (R^2)

NEAR-TERM VALUATION TRENDS ARE
HIGHLY PREDICTABLE USING:

- MOVING AVERAGES
- MOMENTUM
- TRAILING VOLATILITY

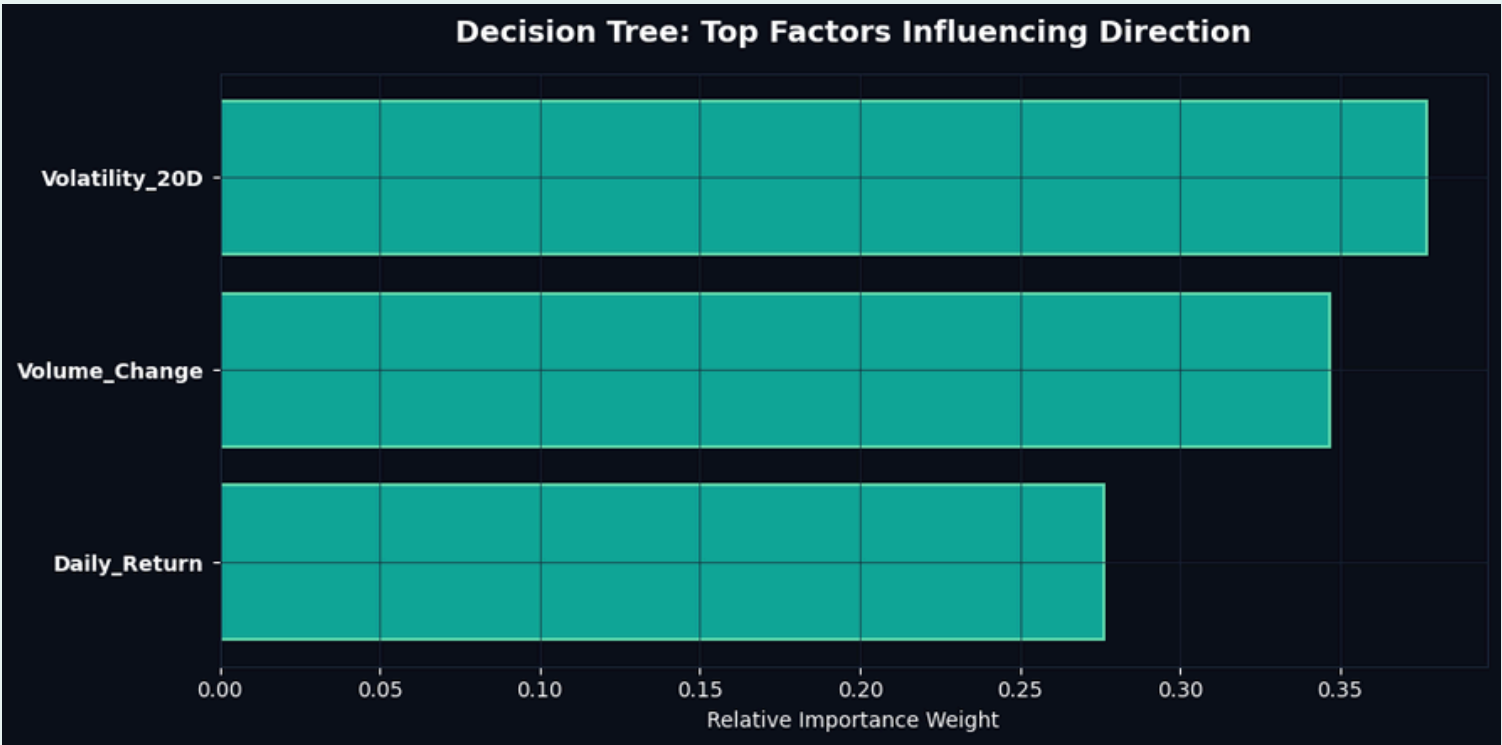
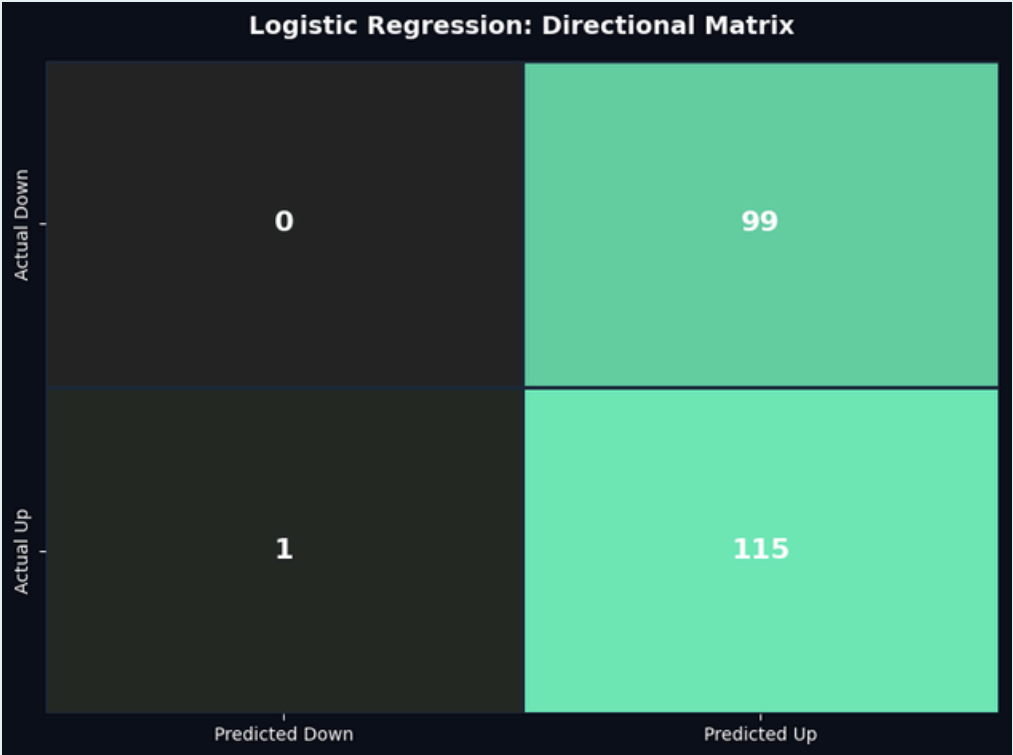
Regression: Distribution of Prediction Errors



DIRECTIONAL CLASSIFICATION

53.5% DIRECTIONAL ACCURACY

THE “BULL BIAS” PROBLEM:
NVDA TRENDS UPWARD SO
CONSISTENTLY THAT THE
MODEL IDENTIFIES BULLISH
MOMENTUM WELL, BUT
STRUGGLES TO PREDICT
SUDDEN CORRECTIONS
FROM TECHNICAL
INDICATORS ALONE.



STRATEGIC TAKEAWAYS

Key insights from machine learning analysis on NVIDIA (NVDA)

Trading Edge

- Regression models identify price zones effectively
- Better for trend forecasting than directional flips
- Strong near-term predictive capability

Forecasting Accuracy

- Regression model achieved 86.6% R^2 accuracy
- Momentum and volatility were key predictors
- Price behavior showed strong statistical consistency

Classification Challenge

- Directional prediction accuracy reached 53.5%
- Bullish market bias reduced downside detection
- Technical indicators alone miss sentiment shocks

Future Evolution

- Integrate NLP sentiment analysis
- Combine technical + news-based signals
- Expand toward real-time AI trading systems